Thermodynamics and Statistical Mechanics — Practice Set 1

Student 2024001

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Problem 1 A system consists of N **distinguishable, independent particles** obeying **Boltzmann statistics**. Each particle can occupy one of three non-degenerate energy levels: $-E$, 0, and E. The system is in thermal contact with a heat reservoir at temperature T.

Define $\beta = \frac{1}{k_B T}$. The single-particle partition function is:

$$z = e^{\beta E} + 1 + e^{-\beta E} = 1 + 2\cosh(\beta E)$$

The occupation probabilities for the three levels are:

$$P(-E) = \frac{e^{\beta E}}{z}, \quad P(0) = \frac{1}{z}, \quad P(E) = \frac{e^{-\beta E}}{z}$$

- **(a)** Find the entropy at $T = 0$ K.
- **(b)** Find the maximum entropy.
- **(c)** Find the minimum energy of the system.
- **(d)** Find the total partition function Z for the N-particle system.
- **(e)** Find the average energy $\langle U \rangle$ of the system.
- **(f)** Evaluate the integral

$$\int_0^\infty \frac{C(T)}{T}\,\mathrm{d}T$$

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(a) Entropy at $T = 0$ K

As $T \to 0$, we have $\beta \to \infty$, so $P(-E) \to 1$ and $P(0), P(E) \to 0$.

Every particle occupies the ground state $-E$. There is only **one microstate** (all N particles on $-E$), so:

$$S = k_B \ln 1 = \boxed{0}$$

This is consistent with the **Third Law of Thermodynamics**.

(b) Maximum Entropy

Maximum entropy corresponds to $T \to \infty$ ($\beta \to 0$), where all three levels are **equally probable**.

For N distinguishable particles, a configuration with occupation numbers (n_1, n_2, n_3) where $n_1 + n_2 + n_3 = N$ has a multiplicity:

$$W = \frac{N!}{n_1!\,n_2!\,n_3!}$$

W is maximized when $n_1 = n_2 = n_3 = \frac{N}{3}$. Using **Stirling's approximation** ($\ln n! \approx n \ln n - n$):

$$\begin{aligned} S_{\max} &= k_B \ln\left(\frac{N!}{\frac{N}{3}!^3}\right) \\ &\approx k_B\left[N\ln N - 3\cdot\frac{N}{3}\cdot\ln\left(\frac{N}{3}\right)\right] \\ &= k_B N \ln 3 \end{aligned}$$

$$\boxed{S_{\max} = \mathrm{N}\ k_B\ \ln\ 3}$$

(c) Minimum Energy

The minimum energy occurs when **all** N particles are in the lowest level $-E$:

$$\boxed{U_{\min} = -\mathrm{N\ E}}$$

(d) Partition Function

Since the N particles are **distinguishable and independent**, the total partition function is:

$$\boxed{\mathrm{Z} = z^N = \left[1 + 2\cosh\left(\frac{E}{k_B T}\right)\right]^N}$$

(e) Average Energy

The average energy of a single particle:

$$\begin{aligned} \langle\varepsilon\rangle &= (-E)\cdot\frac{e^{\beta E}}{z} + (0)\cdot\frac{1}{z} + (E)\cdot\frac{e^{-\beta E}}{z} \\ &= E\frac{e^{-\beta E} - e^{\beta E}}{z} \\ &= -\frac{2E\sinh(\beta E)}{1 + 2\cosh(\beta E)} \end{aligned}$$

For N particles:

$$\boxed{\langle\ \mathrm{U}\ \rangle = -\frac{2NE\sinh\left(\frac{E}{k_B T}\right)}{1+2\cosh\left(\frac{E}{k_B T}\right)}}$$

(f) Integral of $\frac{C(T)}{T}$

We need to evaluate:

$$\int_0^\infty \frac{C(T)}{T}\,\mathrm{d}T = ?$$

Since the energy levels are fixed (no volume degree of freedom), entropy depends only on temperature:

$$\mathrm{d}S = \frac{C(T)}{T}\,\mathrm{d}T$$

Therefore, the integral equals the **total entropy change**:

$$\int_0^\infty \frac{C(T)}{T}\,\mathrm{d}T = \int_0^\infty \mathrm{d}S = S(T=\infty) - S(T=0)$$

Substituting results from parts (a) and (b):

$$\boxed{\int_0^\infty \frac{C(T)}{T}\,\mathrm{d}\ \mathrm{T} = \mathrm{N}\ k_B\ \ln\ 3}$$

Summary of Results

Part	Quantity	Result
(a)	Entropy at $T = 0$	0
(b)	Maximum entropy	$Nk_B \ln 3$
(c)	Minimum energy	$-NE$
(d)	Partition function	$\left[1 + 2\cosh\left(\frac{E}{k_B}T\right)\right]^N$
(e)	Average energy	$-2NE\frac{\sinh\left(\frac{E}{k_B}T\right)}{\left[1+2\cosh\left(\frac{E}{k_B}T\right)\right]}$
(f)	Integral of $\frac{C(T)}{T}$	$Nk_B \ln 3$